Unrestricted Coupled Cluster Singles and Doubles for Complex Orbitals

1 Scope

The standard spin-orbital CCSD equations (Stanton, Gauss, Watts, Bartlett, J. Chem. Phys. **94**, 4334 (1991)) are written for real orbitals. In that form several integrals appear in a particular index order that is only correct because of the real-orbital symmetry $\langle pq\|rs\rangle = \langle rs\|pq\rangle$. Here the equations are re-derived by tracking which operator string every integral multiplies, so that they hold for *arbitrary complex* spin-orbitals $\varphi_p(\mathbf{r},\sigma)$. This covers

- UCCSD with complex spatial orbitals, $\varphi_p = \psi_p(\mathbf{r})\,\sigma_p$ with $\sigma_p \in \{\alpha,\beta\}$ (Section 7);
- general two-component or four-component spinors (the case needed for the relativistic code), where there is no spin factorization at all.

No time-reversal or other symmetry is assumed. The reference determinant $|0\rangle$ is built from the N occupied spin-orbitals and need not be a Hartree–Fock solution (the Fock matrix may have non-zero occupied–virtual blocks).

2 Conventions

Indices i,j,k,l,m,n label occupied, a,b,c,d,e,f virtual, p,q,r,s general spin-orbitals. The one-electron operator is h, with $h_{pq} = \langle p|h|q\rangle = \int \varphi_p^*\, h\, \varphi_q$. Integrals:

$$\langle pq|rs\rangle = \iint \varphi_p^*(1)\,\varphi_q^*(2)\,\frac{1}{r_{12}}\,\varphi_r(1)\,\varphi_s(2), \tag{1}$$

$$\langle pq\|rs\rangle = \langle pq|rs\rangle - \langle pq|sr\rangle. \tag{2}$$

The Hamiltonian in second quantization,

$$H = \sum_{pq} h_{pq}\, a_p^\dagger a_q + \tfrac{1}{4}\sum_{pqrs}\langle pq\|rs\rangle\, a_p^\dagger a_q^\dagger a_s a_r, \tag{3}$$

so that $\langle pq\|rs\rangle$ is always the coefficient of the string $a_p^\dagger a_q^\dagger a_s a_r$: *the two left indices are the created orbitals, the two right indices the annihilated ones.* This single rule fixes every conjugation below.

The Fock matrix and the reference energy are

$$f_{pq} = h_{pq} + \sum_i \langle pi\|qi\rangle, \qquad E_0 = \sum_i h_{ii} + \tfrac{1}{2}\sum_{ij}\langle ij\|ij\rangle, \tag{4}$$

$f_{pq} = f_{qp}^*$ (Hermitian, in general complex).

Integral symmetries

	real orbitals	complex orbitals
$\langle pq\|\|rs\rangle = -\langle qp\|\|rs\rangle = -\langle pq\|\|sr\rangle = \langle qp\|\|sr\rangle$	yes	yes
$\langle pq\|\|rs\rangle = \langle rs\|\|pq\rangle$	yes	**no**
$\langle pq\|\|rs\rangle = \langle rs\|\|pq\rangle^*$	(trivial)	yes
$f_{pq} = f_{qp}$	yes	**no**, $f_{pq} = f_{qp}^*$

The 8-fold permutational symmetry of the real case is reduced to 4-fold for the unantisymmetrized $\langle pq|rs\rangle$ (namely $\langle pq|rs\rangle = \langle qp|sr\rangle = \langle rs|pq\rangle^* = \langle sr|qp\rangle^*$). In particular, the blocks with virtuals in the bra and occupied in the ket (e.g. $\langle ab\|ij\rangle$) are the complex conjugates of the corresponding blocks with occupied in the bra (e.g. $\langle ij\|ab\rangle$), and are not equal to them.

3 Cluster ansatz and amplitude equations

$$|\Psi\rangle = e^T|0\rangle, \qquad T = T_1 + T_2, \tag{5}$$

$$T_1 = \sum_{ia} t_i^a\, a_a^\dagger a_i, \qquad T_2 = \tfrac{1}{4}\sum_{ijab} t_{ij}^{ab}\, a_a^\dagger a_b^\dagger a_j a_i, \tag{6}$$

with t_{ij}^{ab} antisymmetric in (ij) and in (ab). The amplitudes are complex and satisfy no conjugation relation such as $t_{ij}^{ab} = t_{ji}^{ba*}$. The equations are the usual projections of the similarity-transformed Hamiltonian $\bar{H} = e^{-T}He^T$,

$$E_c = \langle 0|\bar{H}|0\rangle - E_0, \qquad \langle \Phi_i^a|\bar{H}|0\rangle = 0, \qquad \langle \Phi_{ij}^{ab}|\bar{H}|0\rangle = 0. \tag{7}$$

Neither the T amplitudes nor the left-hand projectors are ever complex conjugated. Complex orbitals therefore enter *only* through the integrals, and the algebra is identical to the real case term by term; what changes is which of the two members of a conjugate pair $\langle pq\|rs\rangle$, $\langle rs\|pq\rangle$ appears. There is no left-hand state and no Λ equation in what follows.

4 Working equations

4.1 Correlation energy

$$E_c = \sum_{ia} f_{ia}\, t_i^a + \tfrac{1}{4}\sum_{ijab}\langle ij\|ab\rangle\, t_{ij}^{ab} + \tfrac{1}{2}\sum_{ijab}\langle ij\|ab\rangle\, t_i^a t_j^b. \tag{8}$$

For the exact wave function E_c is real. For truncated CCSD no identity guarantees it, so take $\mathrm{Re}\, E_c$ and monitor $\mathrm{Im}\, E_c$ (PySCF's GCCSD warns when it exceeds 10^{-4}).

4.2 Intermediates

With

$$\tilde{\tau}_{ij}^{ab} = t_{ij}^{ab} + \tfrac{1}{2}\left(t_i^a t_j^b - t_i^b t_j^a\right), \qquad \tau_{ij}^{ab} = t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a, \tag{9}$$

and $P(pq)X_{pq} = X_{pq} - X_{qp}$:

$$F_{ae} = (1-\delta_{ae})f_{ae} - \tfrac{1}{2}\sum_m f_{me}\,t^a_m + \sum_{mf} t^f_m \langle ma\|fe\rangle - \tfrac{1}{2}\sum_{mnf} \tilde{\tau}^{af}_{mn}\langle mn\|ef\rangle, \tag{10}$$

$$F_{mi} = (1-\delta_{mi})f_{mi} + \tfrac{1}{2}\sum_e t^e_i f_{me} + \sum_{ne} t^e_n \langle mn\|ie\rangle + \tfrac{1}{2}\sum_{nef} \tilde{\tau}^{ef}_{in}\langle mn\|ef\rangle, \tag{11}$$

$$F_{me} = f_{me} + \sum_{nf} t^f_n \langle mn\|ef\rangle, \tag{12}$$

$$W_{mnij} = \langle mn\|ij\rangle + P(ij)\sum_e t^e_j \langle mn\|ie\rangle + \tfrac{1}{4}\sum_{ef} \tau^{ef}_{ij}\langle mn\|ef\rangle, \tag{13}$$

$$W_{abef} = \langle ab\|ef\rangle - P(ab)\sum_m t^b_m \langle am\|ef\rangle + \tfrac{1}{4}\sum_{mn} \tau^{ab}_{mn}\langle mn\|ef\rangle, \tag{14}$$

$$W_{mbej} = \langle mb\|ej\rangle + \sum_f t^f_j \langle mb\|ef\rangle - \sum_n t^b_n \langle mn\|ej\rangle - \sum_{nf}\left(\tfrac{1}{2}t^{fb}_{jn} + t^f_j t^b_n\right)\langle mn\|ef\rangle. \tag{15}$$

Each of these multiplies, in $\bar{H}$, an operator string whose created indices are the left indices (e.g. F_{ae} multiplies $a^\dagger_a a_e$ and W_{abef} multiplies $a^\dagger_a a^\dagger_b a_f a_e$), and every integral inside them keeps the created-left / annihilated-right pattern of that string. This is why these seven definitions need no conjugate: all integrals appearing there are already in the orientation that the operator algebra delivers.

4.3 T_1 equation

$$t^a_i D^a_i = f_{ai} + \sum_e t^e_i F_{ae} - \sum_m t^a_m F_{mi} + \sum_{me} t^{ae}_{im} F_{me} - \sum_{nf} t^f_n \langle na\|if\rangle \\ - \tfrac{1}{2}\sum_{mef} t^{ef}_{im}\langle ma\|ef\rangle - \tfrac{1}{2}\sum_{men} t^{ae}_{mn}\langle nm\|ei\rangle. \tag{16}$$

4.4 T_2 equation

$$t^{ab}_{ij} D^{ab}_{ij} = \langle ab\|ij\rangle + P(ab)\sum_e t^{ae}_{ij}\Big(F_{be} - \tfrac{1}{2}\sum_m t^b_m F_{me}\Big) - P(ij)\sum_m t^{ab}_{im}\Big(F_{mj} + \tfrac{1}{2}\sum_e t^e_j F_{me}\Big) \\ + \tfrac{1}{2}\sum_{mn}\tau^{ab}_{mn}W_{mnij} + \tfrac{1}{2}\sum_{ef}\tau^{ef}_{ij}W_{abef} + P(ij)P(ab)\sum_{me}\Big(t^{ae}_{im}W_{mbej} - t^e_i t^a_m\langle mb\|ej\rangle\Big) \\ + P(ij)\sum_e t^e_i\langle ab\|ej\rangle - P(ab)\sum_m t^a_m\langle mb\|ij\rangle. \tag{17}$$

The denominators contain the diagonal of the (Hermitian, hence real-diagonal) Fock matrix,

$$D^a_i = f_{ii} - f_{aa}, \qquad D^{ab}_{ij} = f_{ii} + f_{jj} - f_{aa} - f_{bb}. \tag{18}$$

The off-diagonal Fock elements enter through the $(1-\delta)$ terms in F_{ae}, F_{mi}, so no canonical-orbital assumption is made. Level shifting and DIIS act on the complex amplitude vector as in the real case, with the DIIS overlap matrix taken as $\mathbf{e}^\dagger_k\mathbf{e}_l$ and the extrapolation coefficients real.

5 Differences with respect to the real-orbital equations

Compared with the equations as usually printed for real orbitals, only the two inhomogeneous terms change, and they change because the real-orbital formulas are written with the occupied-bra orientation using $\langle pq\|rs\rangle = \langle rs\|pq\rangle$:

term	real-orbital form	complex form (this note)
T_1 inhomogeneous term	f_{ia}	$f_{ai} = f_{ia}^*$
T_2 inhomogeneous term	$\langle ij\|ab\rangle$	$\langle ab\|ij\rangle = \langle ij\|ab\rangle^*$

All other terms, including $P(ij)\, t_i^e \langle ab\|ej\rangle$ and $-P(ab)\, t_m^a \langle mb\|ij\rangle$, are printed with the same index order as in the real case; they are correct for complex orbitals because that order is the created-left / annihilated-right order of the operator algebra. What differs is only how they are obtained from stored arrays: if the code stores the occupied-bra blocks ($\langle ij\|mb\rangle$, $\langle je\|ba\rangle$), then

$$\langle ab\|ej\rangle = \langle je\|ba\rangle^*, \qquad \langle mb\|ij\rangle = \langle ij\|mb\rangle^*, \tag{19}$$

(pair swap $\langle pq\|rs\rangle = \langle qp\|sr\rangle$ combined with Hermitian conjugation) and the conjugate must be taken. For real orbitals every conjugation is trivial and the equations reduce to Stanton's. The same pattern is used in PySCF's `GCCSD` (`pyscf/cc/gccsd.py`, `update_amps`), which can serve as an external reference with complex spin-orbitals.

6 Initial guess

With $f_{ov} = 0$ the first iteration from $T = 0$ gives the MP2 amplitudes

$$t_i^a = \frac{f_{ai}}{D_i^a}, \qquad t_{ij}^{ab} = \frac{\langle ab\|ij\rangle}{D_{ij}^{ab}}, \tag{20}$$

and $E_{\mathrm{MP2}} = \frac{1}{4}\sum \langle ij\|ab\rangle\, t_{ij}^{ab} = \frac{1}{4}\sum |\langle ij\|ab\rangle|^2 / D_{ij}^{ab}$ is real and negative as it should be. Using $\langle ij\|ab\rangle$ instead of $\langle ab\|ij\rangle$ in the numerator would give a complex energy, which is a convenient sanity check on the orientation of the integrals.

7 UCCSD: spin blocks for complex spatial orbitals

For $\varphi_p = \psi_p\, \sigma_p$ with complex spatial ψ_p,

$$\langle pq|rs\rangle = \delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s}\,(pr|qs), \qquad (pr|qs) = \iint \psi_p^*(1)\psi_r(1)\,\frac{1}{r_{12}}\,\psi_q^*(2)\psi_s(2), \tag{21}$$

and h, f conserve spin. Spin conservation then restricts the amplitudes to

- t_i^a: $\sigma_i = \sigma_a$, giving t^α and t^β;
- t_{ij}^{ab}: $\sigma_i + \sigma_j = \sigma_a + \sigma_b$, giving the unique blocks $\alpha\alpha \to \alpha\alpha$, $\beta\beta \to \beta\beta$, and $\alpha\beta \to \alpha\beta$ (t_{iJ}^{aB}; the blocks with a, B swapped follow by antisymmetry).

The equations of the previous section hold verbatim in this block structure: the sums run over the spin-allowed combinations and the integral blocks are $\langle pq\|rs\rangle = (pr|qs) - (ps|qr)$ for equal spins (all four orbitals of one spin, or $\sigma_p = \sigma_r, \sigma_q = \sigma_s$) and $\langle pq\|rs\rangle = (pr|qs)$ for the mixed block

$p, r \in \alpha$, $q, s \in \beta$. The conjugation pattern of the previous section applies unchanged to each block; in particular the $\alpha\beta$ inhomogeneous term is $\langle aB\|iJ\rangle = (ai|BJ) = (ia|JB)^*$, with no exchange part.

For genuine spinors (two- or four-component) φ_p is not an eigenfunction of σ_z, the integrals $\langle pq|rs\rangle$ do not factorize, and the spin-orbital equations of Section 4 are used directly with all index combinations, i.e. as a generalized CCSD (GCCSD) with complex orbitals.

8 Cost and storage

The scaling is that of real CCSD, $O(o^2v^4)$ time and $O(o^2v^2)$ amplitude storage, with complex arithmetic (about $4\times$ the flops and $2\times$ the memory per element, using `zgemm` in the contractions). Because $\langle pq\|rs\rangle$ has only 4-fold symmetry, one cannot store only the unique integrals of one orientation and reuse them for the conjugate orientation by a transposition; the conjugate blocks of the previous section are obtained by conjugating the stored block, never by transposing it alone.

9 Left-hand equations (Λ) and reduced density matrices

Everything above is the right-hand side of the CCSD problem. Densities, and any property that is not the energy, need the left-hand state $\langle 0|(1+\Lambda)e^{-T}$,

$$\Lambda = \sum_{ia} \lambda_i^a \, a_i^\dagger a_a + \tfrac{1}{4} \sum_{ijab} \lambda_{ij}^{ab} \, a_i^\dagger a_j^\dagger a_b a_a, \tag{22}$$

with λ_{ij}^{ab} antisymmetric in (ij) and in (ab). The Λ equations are the stationarity conditions of the Lagrangian

$$\mathcal{L}(t,\lambda) = E_c(t) + \sum_{ia} \lambda_i^a \, R_i^a(t) + \tfrac{1}{4} \sum_{ijab} \lambda_{ij}^{ab} \, R_{ij}^{ab}(t), \qquad R_\mu(t) = \langle \Phi_\mu | \bar{H} | 0 \rangle = n_\mu(t) - D_\mu t_\mu, \tag{23}$$

where n_μ is the numerator of Section 4 ($t_\mu = n_\mu / D_\mu$ is the amplitude equation). $\mathcal{L}$ is holomorphic in t and λ (neither is ever complex conjugated), so $\partial\mathcal{L}/\partial t_\mu = 0$ gives *linear* equations for λ with the same complex integrals as the T equations; at convergence $\mathcal{L} = E_c$. With complex orbitals the only change with respect to the real-orbital formulas is again which member of a conjugate pair of integrals appears. The inhomogeneous terms are f_{ia} and $\langle ij\|ab\rangle$ with the occupied indices in the bra and *not* conjugated (the complex conjugates of the T-equation driving terms); the conjugated integrals $\overline{\langle jk\|ic\rangle} = \langle ic\|jk\rangle$ and $\overline{\langle ja\|cb\rangle} = \langle bc\|aj\rangle$ enter only through W^{ovoo} and W^{vvvo} below. The equations are those of PySCF's `gccsd_lambda.py` (`make_intermediates`, `update_lambda`) in the conventions of this note, written out from the same data that was evaluated with `numpy` and compared with PySCF (Section 9.7).

9.1 T-dependent intermediates

$\tau_{ij}^{ab} = t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a$ as in Section 4. The intermediate F_{kc} below equals F_{me} of Section 4.2. $\mathcal{A}, \mathcal{B}$ are one-body, $\mathcal{C}, \mathcal{D}, \mathcal{E}, \mathcal{G}$ and the four W two-body building blocks; all are independent of λ and built once.

$$\begin{aligned} \mathcal{A}_{ba} = f_{ba} &- \sum_j f_{ja} \, t_j^b \\ &- \sum_{jc} \langle jb\|ac\rangle \, t_j^c + \tfrac{1}{2} \sum_{jkc} \langle jk\|ca\rangle \, \tau_{jk}^{bc} \end{aligned}$$

$$\mathcal{B}_{ij} = f_{ij} + \sum_{b} f_{ib}\, t_j^b - \sum_{kb} \langle ki \| jb \rangle\, t_k^b + \tfrac{1}{2} \sum_{kbc} \langle ik \| bc \rangle\, \tau_{jk}^{bc}$$

$$\mathcal{C}_{ijkl} = \sum_{cd} \langle ij \| cd \rangle\, \tau_{kl}^{cd}$$

$$\mathcal{D}_{jcbk} = \sum_{ld} \langle lj \| db \rangle\, t_{kl}^{cd} + \langle jc \| bk \rangle$$

$$F_{kc} = f_{kc} - \sum_{ld} \langle kl \| dc \rangle\, t_l^d$$

$$\mathcal{E}_{bj} = f_{bj} + \sum_{kc} f_{kc}\, t_{jk}^{bc} + \sum_{kc} F_{kc}\, t_k^b\, t_j^c - \tfrac{1}{2} \sum_{klc} \langle kl \| jc \rangle\, t_{kl}^{bc} + \tfrac{1}{2} \sum_{kcd} \langle kb \| dc \rangle\, t_{jk}^{cd}$$

$$\mathcal{G}_{ck} = \mathcal{E}_{ck} + \sum_{jb} \mathcal{D}_{jcbk}\, t_j^b + \sum_{b} \mathcal{A}_{cb}\, t_k^b - \sum_{j} \mathcal{B}_{jk}\, t_j^c$$

$$W_{ijkl}^{\mathrm{oooo}} = \tfrac{1}{2} \langle ij \| kl \rangle + \tfrac{1}{4}\, \mathcal{C}_{ijkl} + \sum_{c} \langle ij \| kc \rangle\, t_l^c$$

$$W_{jcbk}^{\mathrm{ovvo}} = \mathcal{D}_{jcbk} - \sum_{ld} \langle lj \| db \rangle\, t_l^c\, t_k^d - \sum_{l} \langle lj \| kb \rangle\, t_l^c + \sum_{d} \langle jc \| bd \rangle\, t_k^d$$

$$W_{icjk}^{\mathrm{ovoo}} = \tfrac{1}{4} \sum_{bd} \langle ic \| db \rangle\, \tau_{jk}^{db} + \tfrac{1}{2}\, \overline{\langle jk \| ic \rangle} + \sum_{b} \mathcal{D}_{icbk}\, t_j^b - \sum_{lb} \langle li \| jb \rangle\, t_{kl}^{cb}$$

$$W_{bcak}^{\mathrm{vvvo}} = \sum_{j} \mathcal{D}_{jcak}\, t_j^b + \tfrac{1}{4} \sum_{jl} \langle jl \| ka \rangle\, \tau_{jl}^{bc} - \tfrac{1}{2}\, \overline{\langle ka \| cb \rangle} + \sum_{jd} \langle jb \| ad \rangle\, t_{kj}^{cd}$$

9.2 Λ update

The λ-dependent intermediates are

$$M^{\mathrm{v}}_{ba} = \tfrac{1}{2}\sum_{klc} \lambda^{ca}_{kl}\, t^{cb}_{kl}$$

$$M^{\mathrm{o}}_{ij} = \tfrac{1}{2}\sum_{kcd} \lambda^{cd}_{ki}\, t^{cd}_{kj}$$

$$N^{\mathrm{v}}_{ba} = M^{\mathrm{v}}_{ba} + \sum_{k} \lambda^{a}_{k}\, t^{b}_{k}$$

$$N^{\mathrm{o}}_{ik} = M^{\mathrm{o}}_{ik} + \sum_{c} \lambda^{c}_{i}\, t^{c}_{k}$$

$$S_{ijkl} = \sum_{cd} \lambda^{cd}_{ij}\, \tau^{cd}_{kl}$$

$$U_{ijck} = \sum_{d} \lambda^{cd}_{ij}\, t^{d}_{k}$$

$$\begin{aligned} M_{ijab} = {} & \sum_{kl} \lambda^{ab}_{kl}\, W^{\mathrm{oooo}}_{ijkl} + \tfrac{1}{4}\sum_{kl} \langle kl\|ab\rangle\, S_{ijkl} \\ & - \sum_{kc} \langle kc\|ba\rangle\, U_{ijck} + \tfrac{1}{2}\sum_{cd} \lambda^{cd}_{ij}\, \langle cd\|ab\rangle \end{aligned}$$

$$\begin{aligned} q_{jb} = {} & t^{b}_{j} + \sum_{kc} \lambda^{c}_{k}\, t^{cb}_{kj} \\ & - \sum_{d} N^{\mathrm{v}}_{bd}\, t^{d}_{j} - \sum_{l} M^{\mathrm{o}}_{lj}\, t^{b}_{l} \end{aligned}$$

The term $\frac{1}{2}\sum_{cd}\lambda^{cd}_{ij}\langle cd\|ab\rangle$ in M_{ijab} is the Λ-equation particle–particle ladder: like the T-equation ladder it needs only the $vvvv$ slabs, and $\sum_c \langle kc\|ba\rangle\, U_{ijck}$ needs the $ovvv$ slabs, so the streamed scheme of Section 10 applies unchanged. The diagonal Fock elements are moved to the denominators, $\mathcal{A}'_{ba} = \mathcal{A}_{ba} - \delta_{ba} f_{aa}$ and $\mathcal{B}'_{ij} = \mathcal{B}_{ij} - \delta_{ij} f_{ii}$ (they are not part of the right-hand sides). With $P(ij)X_{ijab} = X_{ijab} - X_{jiab}$ and $P(ab)X_{ijab} = X_{ijab} - X_{ijba}$ the numerator of the doubles equation is

$$\mathcal{N}^{ab}_{ij} = \langle ij\|ab\rangle + M_{ijab} + P(ij)P(ab)\, X_{ijab} - P(ab)\, Y_{ijab} + P(ij)\, Z_{ijab}, \tag{24}$$

with

$$X_{ijab} = \lambda^{a}_{i}\, F_{jb} + \sum_{kc} \lambda^{ca}_{ki}\, W^{\mathrm{ovvo}}_{jcbk}$$

$$Y_{ijab} = \sum_k \lambda^a_k \langle ij\|kb\rangle + \sum_c \lambda^{ca}_{ij}\, \mathcal{A}'_{cb} + \sum_c N^{\mathrm{v}}_{ca} \langle ij\|cb\rangle$$

$$Z_{ijab} = \sum_c \lambda^c_j \langle ic\|ab\rangle + \sum_k \lambda^{ab}_{ki}\, \mathcal{B}'_{jk} - \sum_k N^{\mathrm{o}}_{ik} \langle kj\|ab\rangle$$

and that of the singles equation

$$\begin{aligned}\mathcal{N}^a_i = f_{ia} &+ \sum_{jb} \lambda^b_j \langle ib\|aj\rangle \\ &+ \sum_b \lambda^b_i\, \mathcal{A}'_{ba} - \sum_j \lambda^a_j\, \mathcal{B}'_{ij} \\ &- \sum_{jkc} \lambda^{ca}_{kj}\, W^{\mathrm{ovoo}}_{icjk} - \sum_{kbc} \lambda^{bc}_{ik}\, W^{\mathrm{vvvo}}_{bcak} \\ &+ \sum_{jb} M_{ijab}\, t^b_j + \sum_{jb} \lambda^{ba}_{ji}\, \mathcal{G}_{bj} \\ &+ \sum_{jb} \langle ji\|ba\rangle\, q_{jb} + \sum_{bc} \langle ic\|ab\rangle\, N^{\mathrm{v}}_{bc} \\ &- \sum_{jk} \langle ji\|ka\rangle\, N^{\mathrm{o}}_{kj} - \sum_k M^{\mathrm{o}}_{ik}\, F_{ka} \\ &- \sum_c M^{\mathrm{v}}_{ca}\, F_{ic}\end{aligned}$$

Both are divided by the same denominators as the T equations,

$$\lambda^a_i \leftarrow \frac{\mathcal{N}^a_i}{D^a_i}, \qquad \lambda^{ab}_{ij} \leftarrow \frac{\mathcal{N}^{ab}_{ij}}{D^{ab}_{ij}}, \qquad D^a_i = f_{ii} - f_{aa}, \quad D^{ab}_{ij} = f_{ii} + f_{jj} - f_{aa} - f_{bb}, \tag{25}$$

iterated from $\lambda = t$ with DIIS and an optional level shift exactly like the amplitudes. The cost per iteration is that of one T iteration (the same contractions, with λ replacing t in the ladder).

9.3 One-particle density matrix

Let $\gamma_{pq} = \langle 0|(1+\Lambda)e^{-T} a^\dagger_p a_q e^T|0\rangle$, so that the one-body energy is $\sum_{pq} h_{pq}\gamma_{pq}$ (orbital relaxation is not included). In the occupied–virtual partitioning $\gamma_{pq} = \delta_{pq} n_p + D_{qp}$, with the blocks D^{oo}_{ij}, D^{vv}_{ab}, D^{vo}_{ai} and $D^{\mathrm{ov}}_{ia} = \lambda^a_i$ given by

$$D^{\mathrm{oo}}_{ij} = -\sum_e \lambda^e_i\, t^e_j - \tfrac{1}{2} \sum_{mef} \lambda^{ef}_{im}\, t^{ef}_{jm}$$

$$D^{\mathrm{vv}}_{ab} = \sum_m t^a_m\, \lambda^b_m + \tfrac{1}{2} \sum_{mne} t^{ea}_{mn}\, \lambda^{eb}_{mn}$$

$$x^{(1)}_{mi} = \tfrac{1}{2} \sum_{nef} \lambda^{ef}_{mn}\, t^{ef}_{in}$$

$$x^{(2)}_{ae} = \tfrac{1}{2} \sum_{mnf} t^{fa}_{mn}\, \lambda^{fe}_{mn} + \sum_{m} t^{a}_{m}\, \lambda^{e}_{m}$$

$$D^{\mathrm{vo}}_{ai} = \sum_{me} t^{ae}_{im}\, \lambda^{e}_{m} - \sum_{m} x^{(1)}_{mi}\, t^{a}_{m} - \sum_{e} t^{e}_{i}\, x^{(2)}_{ae} + t^{a}_{i}$$

The coupled-cluster (Lagrangian) density is not Hermitian, already for real orbitals (e.g. D^{ov} and D^{vo} differ); the Hermitian part, which is what the real part of the energy and of its derivatives involves, is

$$\bar{\gamma}_{pq} = \tfrac{1}{2}\big(\gamma_{pq} + \gamma^{*}_{qp}\big), \qquad \mathrm{Re}\sum_{pq} h_{pq}\gamma_{pq} = \sum_{pq} h_{pq}\bar{\gamma}_{pq}. \tag{26}$$

PySCF returns $\bar{\gamma}^{\mathsf{T}}$ (its `dm1[p,q]` is $\bar{\gamma}_{qp} = \langle a^{\dagger}_{q} a_{p}\rangle$).

9.4 Two-particle density matrix

Let $\Gamma_{pqrs} = \langle 0|(1+\Lambda)e^{-T} a^{\dagger}_{p} a^{\dagger}_{q} a_{s} a_{r} e^{T}|0\rangle$, so the energy is

$$E = \sum_{pq} h_{pq}\bar{\gamma}_{pq} + \tfrac{1}{4}\sum_{pqrs} \langle pq\|rs\rangle\, \Gamma_{pqrs} \quad \text{(real part)}, \tag{27}$$

and Γ is antisymmetric in (pq) and in (rs), with $\Gamma^{*}_{pqrs} = \Gamma_{rspq}$ and $\sum_{p}\Gamma_{pqps} = (N-1)\bar{\gamma}_{qs}$. It is assembled in four steps; the notation is that of `gccsd_rdm.py`. In this part (and only here) the intermediate $\bar{\tau}^{ab}_{ij} = t^{ab}_{ij} + 2t^{a}_{i}t^{b}_{j}$ is *not* antisymmetrized: the antisymmetrization is carried out by the d-blocks below, and using the antisymmetrized τ in these formulas changes the result.

Intermediates.

$$\bar{\tau}^{ab}_{ij} = t^{ab}_{ij} + 2\, t^{a}_{i}\, t^{b}_{j}$$

$$m_{iajb} = \sum_{kc} \lambda^{ac}_{ik}\, t^{cb}_{kj}$$

$$u^{(1)}_{ia} = \sum_{kc} \lambda^{c}_{k}\, t^{ca}_{ki}$$

$$u^{(2)}_{cb} = \sum_{k} \lambda^{c}_{k}\, t^{b}_{k}$$

$$u^{(3)}_{kj} = \sum_{c} \lambda^{c}_{k}\, t^{c}_{j}$$

$$u^{(4)}_{lj} = \sum_{d} m_{ldjd}$$

$$u^{(5)}_{db} = \sum_{l} m_{ldlb}$$

$$u^{(6)}_{ijkl} = \tfrac{1}{16} \sum_{cd} \lambda^{cd}_{kl}\, \bar{\tau}^{cd}_{ij}$$

$$u^{(7)}_{ij} = \tfrac{1}{4} \sum_{c} m_{icjc}$$

$$u^{(8)}_{ab} = \tfrac{1}{4} \sum_{k} m_{kakb}$$

***g*-blocks.** The *oovv*, *ovvv* and *ooov* blocks are complex conjugated before the last (unconjugated) term of the last two is added:

$$\begin{aligned} g^{\mathrm{oovv}}_{ijab} = \Big[& \tfrac{1}{4}\, \overline{\lambda^{ab}_{ij}} + \tfrac{1}{4}\, \bar{\tau}^{ab}_{ij} \\ & + u^{(1)}_{ia}\, t^{b}_{j} + \tfrac{1}{2} \sum_{c} u^{(2)}_{cb}\, t^{ca}_{ij} \\ & + \tfrac{1}{2} \sum_{k} \bar{\tau}^{ab}_{ki}\, u^{(3)}_{kj} - \tfrac{1}{4} \sum_{l} u^{(4)}_{lj}\, \bar{\tau}^{ba}_{li} \\ & - \tfrac{1}{4} \sum_{d} u^{(5)}_{db}\, \bar{\tau}^{da}_{ji} - \tfrac{1}{2} \sum_{ld} m_{ldia}\, \bar{\tau}^{bd}_{lj} \\ & + \sum_{kl} u^{(6)}_{ijkl}\, \bar{\tau}^{ab}_{kl} \Big]^{*} \end{aligned}$$

$$g^{\mathrm{vvvv}}_{abcd} = \tfrac{1}{8} \sum_{ij} \bar{\tau}^{ab}_{ij}\, \lambda^{cd}_{ij}$$

$$g^{\mathrm{oooo}}_{klij} = \tfrac{1}{8} \sum_{ab} \lambda^{ab}_{kl}\, \bar{\tau}^{ab}_{ij}$$

$$\begin{aligned} g^{\mathrm{ooov}}_{jkia} = \Big[& - \tfrac{1}{4} \sum_{b} \bar{\tau}^{ba}_{jk}\, \lambda^{b}_{i} + \sum_{l} g^{\mathrm{oooo}}_{iljk}\, t^{a}_{l} \\ & - u^{(7)}_{ij}\, t^{a}_{k} + \tfrac{1}{2} \sum_{c} m_{icja}\, t^{c}_{k} \Big]^{*} + \tfrac{1}{4} \sum_{b} \lambda^{ab}_{jk}\, t^{b}_{i} \end{aligned}$$

$$\begin{aligned} g^{\mathrm{ovvo}}_{ibaj} = {} & \lambda^{a}_{i}\, t^{b}_{j} + m_{iajb} \\ & - \sum_{kc} \lambda^{ac}_{ik}\, t^{c}_{j}\, t^{b}_{k} \end{aligned}$$

$$g^{\mathrm{ovvv}}_{iabc} = \Big[\tfrac{1}{4}\sum_j \lambda^a_j\,\bar{\tau}^{bc}_{ij} + \sum_d g^{\mathrm{vvvv}}_{bcad}\,t^d_i + u^{(8)}_{ac}\,t^b_i + \tfrac{1}{2}\sum_k m_{kaib}\,t^c_k\Big]^* + \tfrac{1}{4}\sum_j \lambda^{bc}_{ij}\,t^a_j$$

Here g^{oooo} in g^{ooov} is the unconjugated block, and the g^{ovvo} and g^{vvvv} blocks enter the d-blocks without conjugation.

d-blocks. Antisymmetrizing and symmetrizing the g-blocks,

$$\begin{aligned}
d^{\mathrm{ovov}}_{iajb} &= g^{\mathrm{oovv}}_{ijab} - g^{\mathrm{oovv}}_{ijba}\\
d^{\mathrm{vvvv}}_{abcd} &= g^{\mathrm{vvvv}}_{acbd} - g^{\mathrm{vvvv}}_{acdb}\\
d^{\mathrm{oooo}}_{ijkl} &= g^{\mathrm{oooo}}_{ikjl} - g^{\mathrm{oooo}}_{iklj}\\
d^{\mathrm{ovvv}}_{iabc} &= g^{\mathrm{ovvv}}_{ibac} - g^{\mathrm{ovvv}}_{ibca}\\
d^{\mathrm{ooov}}_{ijka} &= g^{\mathrm{ooov}}_{ikja} - g^{\mathrm{ooov}}_{kija}\\
d^{\mathrm{ovvo}}_{iabj} &= g^{\mathrm{ovvo}}_{ibaj}
\end{aligned}$$

and then

$$\begin{aligned}
&d^{\mathrm{ovov}}_{iajb} \leftarrow \tfrac{1}{2}\big(d^{\mathrm{ovov}}_{iajb} + d^{\mathrm{ovov}}_{jbia}\big),\\
&d^{\mathrm{vvvv}}_{abcd} \leftarrow d^{\mathrm{vvvv}}_{abcd} + \overline{d^{\mathrm{vvvv}}_{badc}}, \qquad d^{\mathrm{oooo}}_{ijkl} \leftarrow d^{\mathrm{oooo}}_{ijkl} + \overline{d^{\mathrm{oooo}}_{jilk}},\\
&d^{\mathrm{ovvo}}_{iabj} \leftarrow \tfrac{1}{2}\big(d^{\mathrm{ovvo}}_{iabj} + \overline{d^{\mathrm{ovvo}}_{jbai}}\big).
\end{aligned}$$

Assembly. Define the array $\mathcal{R}_{pqrs} \equiv \Gamma_{qspr}$ blockwise (i,j,k occupied; a,b,c virtual),

$$\begin{aligned}
\mathcal{R}_{iajb} &= d^{\mathrm{ovov}}_{iajb}\\
\mathcal{R}_{aibj} &= \overline{d^{\mathrm{ovov}}_{iajb}}\\
\mathcal{R}_{ijab} &= -d^{\mathrm{ovvo}}_{ibaj}\\
\mathcal{R}_{abij} &= -d^{\mathrm{ovvo}}_{ibaj}\\
\mathcal{R}_{iabj} &= d^{\mathrm{ovvo}}_{iabj}\\
\mathcal{R}_{aijb} &= \overline{d^{\mathrm{ovvo}}_{iabj}}\\
\mathcal{R}_{abcd} &= d^{\mathrm{vvvv}}_{abcd}\\
\mathcal{R}_{ijkl} &= d^{\mathrm{oooo}}_{ijkl}\\
\mathcal{R}_{iabc} &= d^{\mathrm{ovvv}}_{iabc}\\
\mathcal{R}_{abic} &= d^{\mathrm{ovvv}}_{icab}\\
\mathcal{R}_{abci} &= \overline{d^{\mathrm{ovvv}}_{icba}}\\
\mathcal{R}_{aibc} &= \overline{d^{\mathrm{ovvv}}_{iacb}}\\
\mathcal{R}_{ijka} &= d^{\mathrm{ooov}}_{ijka}\\
\mathcal{R}_{iajk} &= d^{\mathrm{ooov}}_{jkia}\\
\mathcal{R}_{ijak} &= \overline{d^{\mathrm{ooov}}_{jika}}\\
\mathcal{R}_{aijk} &= \overline{d^{\mathrm{ooov}}_{kjia}}
\end{aligned}$$

where the block of $\mathcal{R}$ is selected by the occupied/virtual character of p, q, r, s in that order. The correlation part of the one-particle matrix, $\bar{\delta}_{pq} \equiv \bar{\gamma}_{qp} - \delta_{pq} n_p$ (with $n_p = 1$ for occupied p), is then added for every occupied i,

$$\mathcal{R}_{ii\,pq} \mathrel{+}= \bar{\delta}_{pq}, \quad \mathcal{R}_{pq\,ii} \mathrel{+}= \bar{\delta}_{pq}, \quad \mathcal{R}_{p\,ii\,q} \mathrel{-}= \bar{\delta}_{pq}, \quad \mathcal{R}_{i\,pq\,i} \mathrel{-}= \bar{\delta}_{qp}, \tag{28}$$

followed by the mean-field part, $\mathcal{R}_{iijj} \mathrel{+}= 1$ and $\mathcal{R}_{ijji} \mathrel{-}= 1$ for all occupied i, j, and finally

$$\Gamma_{pqrs} = \mathcal{R}_{rpsq}. \tag{29}$$

(For real orbitals every conjugation is trivial and Γ reduces to the usual 2-RDM of spin-orbital CCSD.)

9.5 Storage

Of the g/d-blocks only g^{vvvv} is $O(v^4)$. To avoid ever storing it, contract it directly with the integrals: $\frac{1}{4}\sum\langle ab\|cd\rangle\,\Gamma_{abcd}$ with $\Gamma_{abcd} = d^{\mathrm{vvvv}}_{cadb}$ and $g^{\mathrm{vvvv}}_{abcd} = \frac{1}{8}\sum_{ij} \bar{\tau}^{ab}_{ij}\lambda^{cd}_{ij}$ is evaluated slab by slab exactly like the T ladder, $O(o^2v^4)$ time and no v^4 memory. Any other use of Γ_{abcd} needs it explicitly.

9.6 Density matrix files (`CC_RDM`)

`src/CC/lambda_Uccsd.{h,cpp}` solves the Λ equations above (Jacobi sweeps with DIIS from $\lambda = t$, the same convergence criterion as T; the $vvvv$ ladder is streamed one slab $\langle c\,d|a\,b\rangle$ at a time, the $ovvv$ block $\langle ma\|ef\rangle$ is held once) and writes

- `<input>.dm1`: records (`int32` p, q; `float64` Re, Im) with $\bar{\gamma}_{pq} = \langle a_p^\dagger a_q\rangle$ (Hermitian part), all p, q;
- `<input>.dm2`: records (`int32` p, q, r, s; `float64` Re, Im) with $\Gamma_{pqrs} = \langle a_p^\dagger a_q^\dagger a_s a_r\rangle$ for $p < q$, $r < s$ only (the remaining elements follow from $\Gamma_{pqrs} = -\Gamma_{qprs} = -\Gamma_{pqsr}$); the energy is $E = \sum_{pq} h_{pq}\bar{\gamma}_{pq} + \frac{1}{2}\sum_{pqrs}\langle pq|rs\rangle\Gamma_{pqrs}$;

both ended by an all-zero record. Indices are 1-based and in Kramers-pair order, $(1, 2) = (\psi_1, \bar{\psi}_1)$, $(3, 4) = (\psi_2, \bar{\psi}_2), \ldots$: X2C and the positive-energy block of the 4-component run are numbered that way already, the non-relativistic spin-orbitals are renumbered $\alpha_k \to 2k + 1$, $\beta_k \to 2k + 2$. Frozen-core spinors are included with their mean-field contributions (they appear only through the one-body and mean-field terms of Section 9.4). Elements with $|\Gamma|$ below `CC_RDM_THRESHOLD` are omitted, and the energy recomputed from the written elements is printed next to $E_0 + E_c$ as a check (the two agree to 10^{-9} Eh for the default 10^{-10}). The 2-RDM is produced element by element from the d-blocks (never as a dense n^4 array); only the all-virtual block is formed slab by slab from $\bar{\tau}$ and λ. The densities are unrelaxed: no orbital response is included.

9.7 Numerical check

PySCF 2.14 (`GCCSD`) is the reference. All checks use complex integrals: a Hartree–Fock reference of LiH is rotated inside the occupied and virtual spaces by random unitaries e^A, $A = -A^\dagger$ complex (and, in a second set, with a small complex occupied–virtual mixing so that $f_{ov} \neq 0$ and E_c acquires a small imaginary part).

- The equations above, evaluated from the same data, reproduce PySCF's intermediates and one full Λ update to 10^{-15}, and the d-blocks and $\mathcal{R}$ array to 10^{-16}.

- $\partial\mathcal{L}/\partial t_\mu$ at PySCF's converged (t, λ) vanishes to 3×10^{-12} for all singles and all antisymmetry-unique doubles.
- Central finite differences of $\mathrm{Re}\,E_{\mathrm{CC}}$ (amplitudes re-solved) with respect to a Hermitian perturbation of h and of the antisymmetrized integrals equal $\sum h'_{pq}\bar{\gamma}_{pq}$ and $\frac{1}{4}\sum\langle pq\|rs\rangle'\Gamma_{pqrs}$ to 10^{-8}.
- $\sum h\bar{\gamma} + \frac{1}{4}\sum\langle pq\|rs\rangle\Gamma_{pqrs} = \mathrm{Re}(E_0 + E_c)$ to 10^{-14}.

10 Implementation (`src/CC/Uccsd.{h,cpp}`)

The code is a transcription of Section 4 with two reorganizations that do not change the equations. Every contraction is a two-operand einsum evaluated as one `zgemm`.

- **No stored $\langle ma\|ef\rangle$ or $\langle ab\|ef\rangle$ blocks.** The integrals are supplied as a Coulomb factorization $(pq|rs) = \sum_P \mathrm{bra}_P(p,q)\,\mathrm{ket}_P(r,s)$ (RI: bra = ket = B_P; Cholesky: $\mathrm{bra}_L(a,c) = V_L(c,a)$, $\mathrm{ket}_L(b,d) = V_L^*(b,d)$), and one slab with a fixed first index, $\langle p\,a|ef\rangle$, is rebuilt by a single `zgemm` whenever it is needed. A loop over occupied m accumulates every term containing $\langle ma\|ef\rangle$ (F_{ae}, the T_1 equation, W_{mbej}, the $P(ij)\,t_i^e\langle ab\|ej\rangle$ term and the intermediate Z below); a loop over virtual a applies the particle–particle ladder. Memory is $O(o^2v^2 + N_{\mathrm{aux}}n^2)$, plus $O(N_{\mathrm{aux}}v^2)$ for the cached second factor. For dense integrals the same slabs are read element by element.
- **The ladder is applied as $\sum_{ef}\tau_{ij}^{ef}\langle ab|ef\rangle$.** Because τ is antisymmetric in (ef), $\frac{1}{2}\sum_{ef}\tau_{ij}^{ef}\langle ab\|ef\rangle = \sum_{ef}\tau_{ij}^{ef}\langle ab|ef\rangle$, so no antisymmetrized $vvvv$ slab is formed. The two other pieces of $\frac{1}{2}\sum\tau W_{abef}$ are $-P(ab)\sum_m t_m^b Z_{ijm}^a$ with $Z_{ijm}^a = \frac{1}{2}\sum_{ef}\tau_{ij}^{ef}\langle am\|ef\rangle$, and the $\tau\tau$ term, which is moved into W_{mnij} by doubling its coefficient there ($\frac{1}{4}\to\frac{1}{2}$): $\frac{1}{2}\cdot\frac{1}{4} + \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{2}\cdot\frac{1}{2}$.

Amplitudes are updated by Jacobi sweeps, $t \leftarrow (\mathrm{RHS} - \sigma t)/(D - \sigma)$ with level shift σ, accelerated by DIIS on the antisymmetry-unique amplitudes (real weights, `Utils/DIIS.h`). The Fock matrix is built from the integrals (frozen occupied spinors included), so $f_{ov} \neq 0$ and non-canonical orbitals are handled. Checked against PySCF 2.14: LiH/6-31G, water/cc-pVDZ with frozen core and CO/cc-pVDZ with frozen core (dense and Cholesky agree to 10^{-10}–10^{-8} Eh with PySCF's RCCSD), and the $c\to\infty$ limit of the X2C and 4-component references reproduces the non-relativistic result.